```matlab
clc;
clear;
close all;

%% Signal Parameters
Fs = 1000;
N = 1024;
t = (0:N-1)/Fs;

%% Generate Signal
x = sin(2*pi*100*t) + 0.5*sin(2*pi*200*t);

%% Add White Gaussian Noise
xn = x + 0.5*randn(size(x));

%% Time Domain Signal
figure;
plot(t,xn);
grid on;
title('Noisy Signal');
xlabel('Time (s)');
ylabel('Amplitude');

%% Periodogram PSD
figure;
periodogram(xn,[],[],Fs);
title('Power Spectral Density using Periodogra

%% Signal and Noise Power
signal_power = bandpower(x,Fs,[0 Fs/2]);
noise = xn - x;
noise_power = bandpower(noise,Fs,[0 Fs/2]);

fprintf('Signal Power = %f\n',signal_power);
fprintf('Noise Power = %f\n',noise_power);
```

Figures

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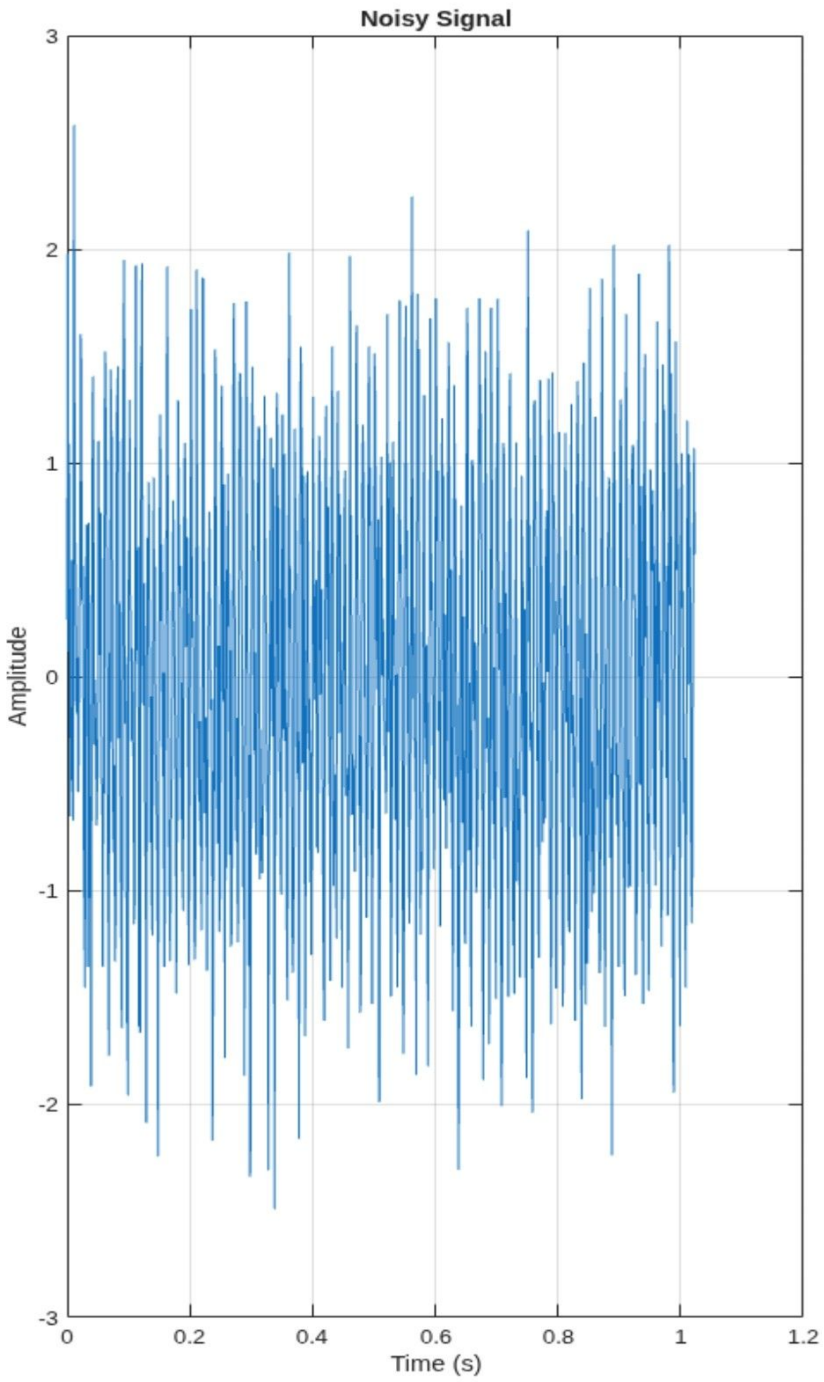
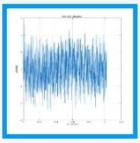
Figure 1: dsp7_1





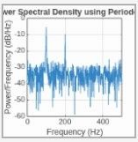


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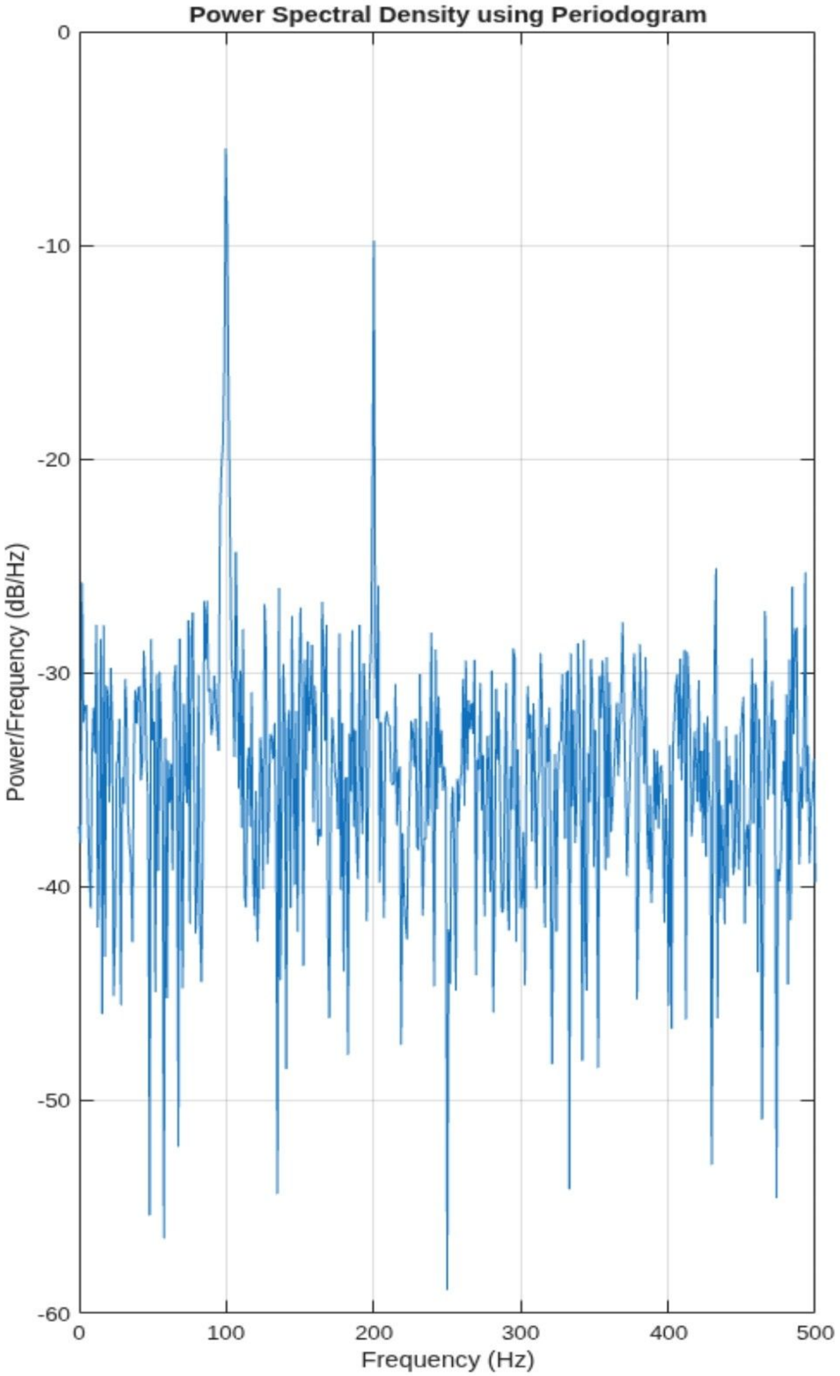
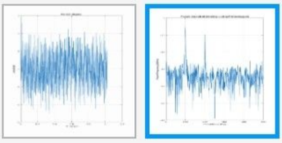
Figure 2: dsp7_1





```
clc;
clear;
close all;

% Signal
fs = 1000;
N = 256;
t = (0:N-1)/fs;
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);

NFFT = 2048;
f = (0:NFFT/2-1)*(fs/NFFT);

% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};

% Processing
for i = 1:3
    xw = x(:).*w{i};
    X = fft(xw,NFFT);
    P{i} = abs(X(1:NFFT/2));
    P{i} = P{i}/max(P{i});
end

%% Individual Spectrums
figure;
for i = 1:3
    subplot(3,1,i);
    plot(f,20*log10(P{i}),'LineWidth',1.5);
    title(['Spectrum using ' names{i} ' Window
    xlim([0 250]);
    grid on;
end

%% Comparison
figure;
plot(f,20*log10(P{1}),'k', ...
     f,20*log10(P{2}),'r', ...
     f,20*log10(P{3}),'b','LineWidth',1.5);
legend(names);
title('Window Comparison');
xlim([50 170]);
grid on;

%% Window Shapes
figure;
```

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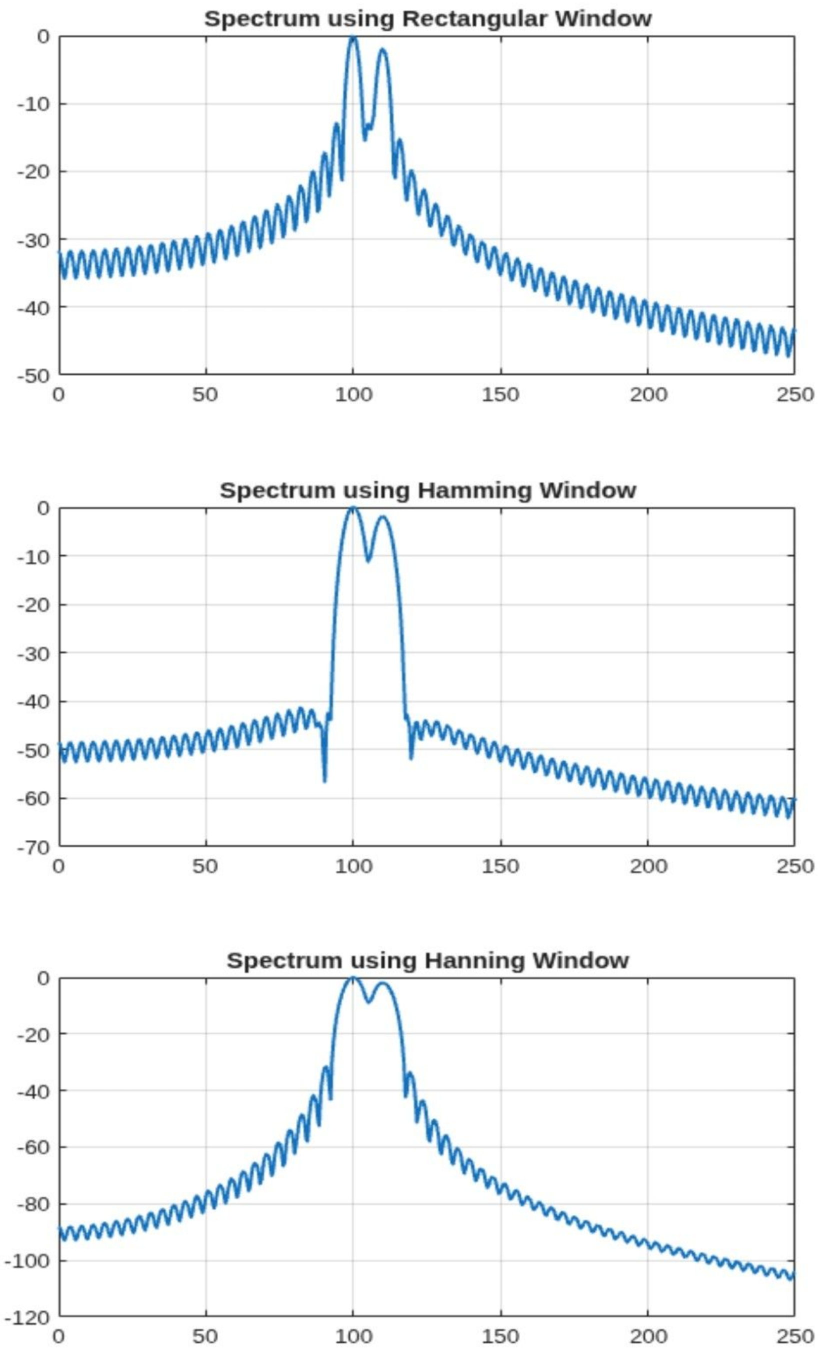
Figure 1: dsp7_2



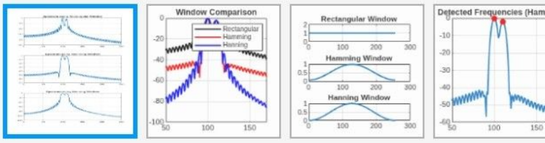
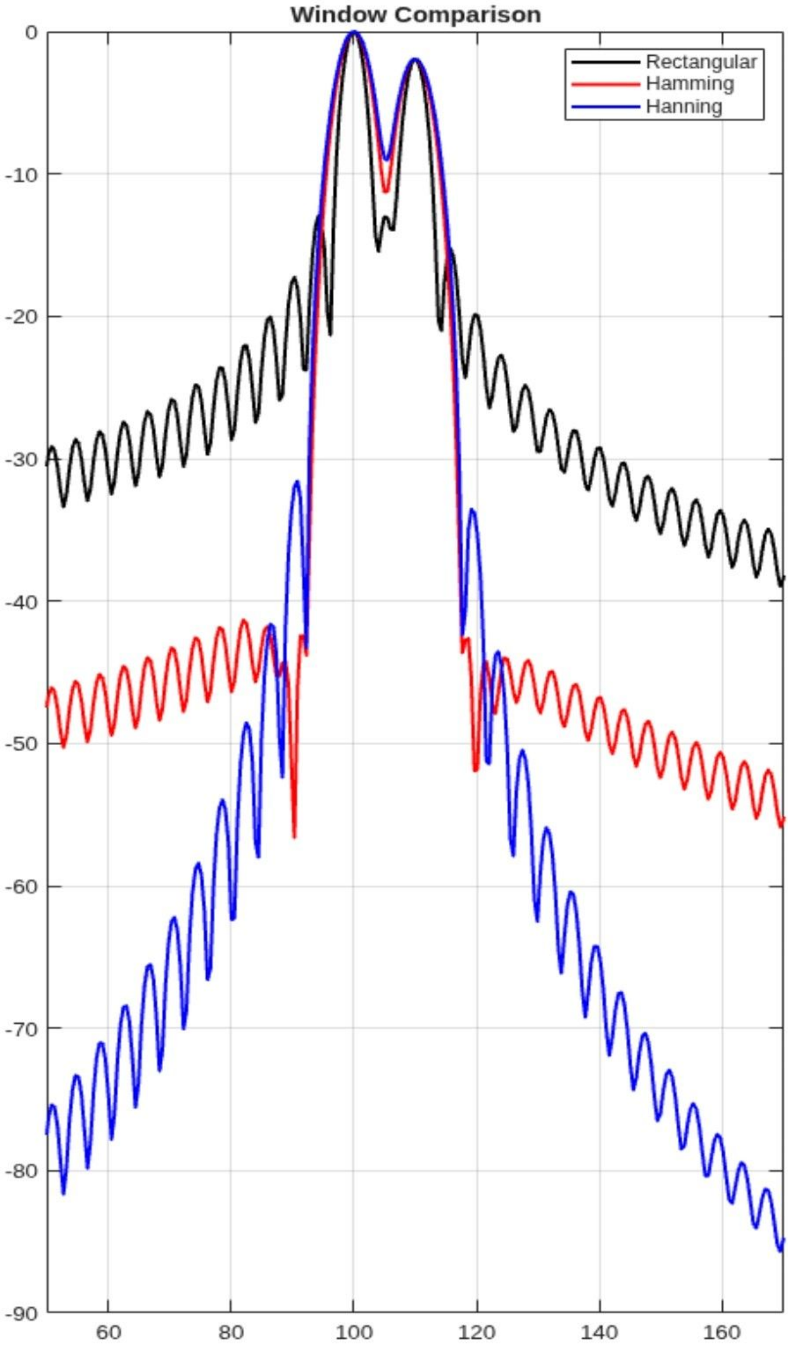


Figure 2: dsp7_2



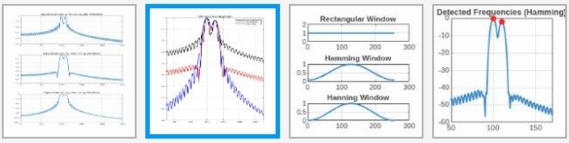


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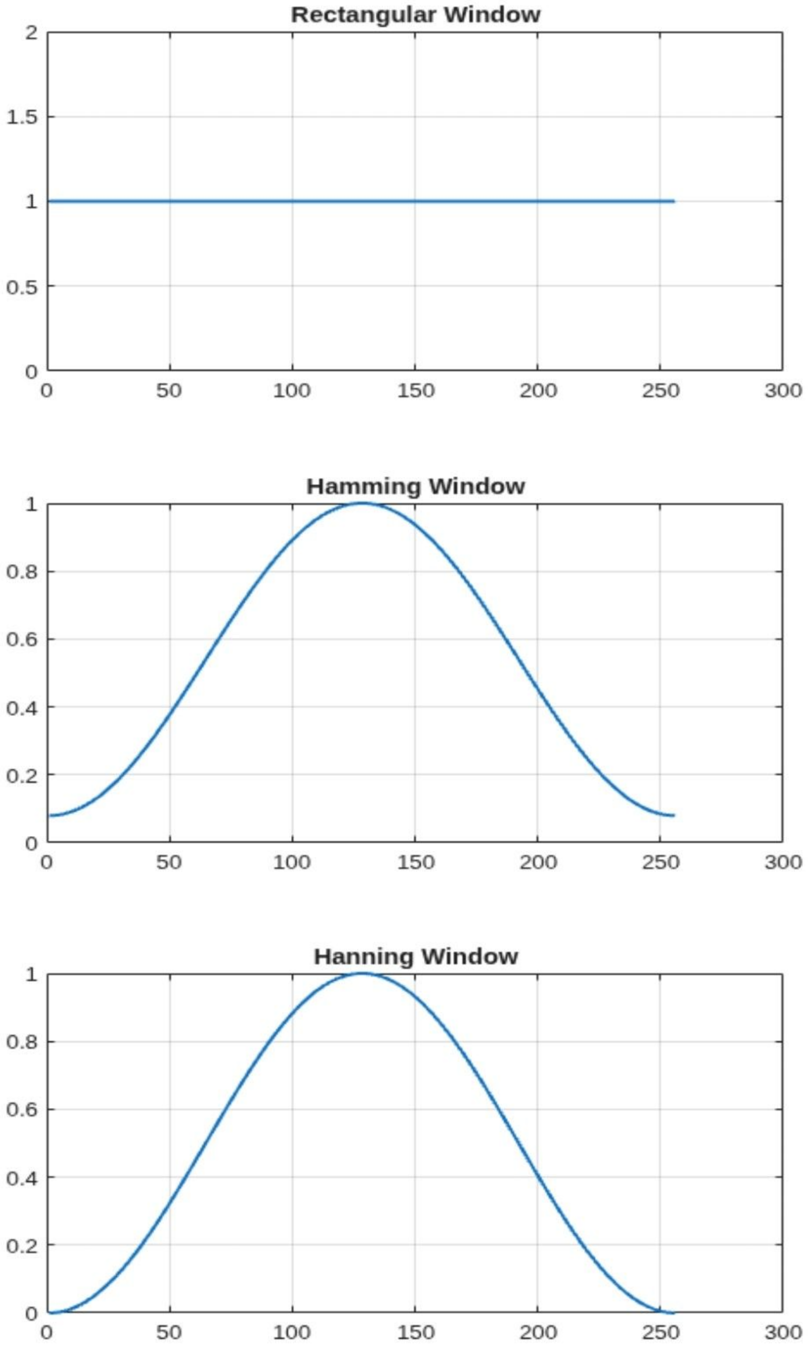
Figure 3: dsp7_2



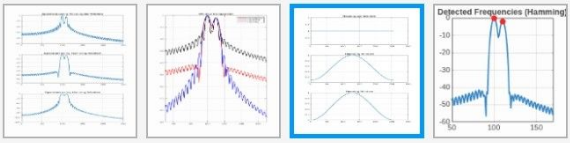


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Figure 4: dsp7_2